Don’t just ship the container

Lessons from 42 TiB of low-level CMS open data

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What we did

I have been on both sides of CMS open data: user and co-producer. Our end-to-end study (Phys. Rev. D 105, 052008) classified boosted top-quark jets from low-level detector data, including tracker hits that standard CMS open data lacks. With the preserved 2012 software (CMSSW_5_3_32), we generated, simulated and reconstructed our own top-quark and QCD samples, keeping the hits.

Under CC0 we released the hit-enriched simulation (30.6 TiB) and a ROOT ntuple of its 9.4 million events (11.7 TiB). Keeping the hits made one study’s samples about 2% of all CMS open data in 2019. The production tool and generation examples went out under GPL v3.

What went wrong: two jobs, one environment

Open data has two jobs: preservation, so that results stay reproducible, and reuse, so that outsiders, including computer scientists building ML, can use the data easily.

The 2012 environment did the first job well: years later it still produced new full simulation. But a container or virtual machine freezes an old operating system and an old Python that modern ML libraries do not support. No modern work happens inside it: we extracted the data inside and trained outside, a bridge each new study must rebuild.

What a release has to ship with

In my experience, CMS releases meet the bar for preservation, but not yet for reuse. They ship the software, calibrations, good-run lists and documentation that keep results reproducible. What they lack, and what I wish we had been given, is either:

1. **A portable copy:** the data in a widely supported format such as HDF5, or a timeless one such as compressed CSV.
2. **A byte-level specification:** not only which collections a file holds and what they mean, but how to decode its bytes, complete enough that an LLM could write a working reader in whatever language is current 20 years from now.

In hindsight, our own ntuple falls short too: it needs no CMS software, but its documentation covers only variable names and meanings, leaving the byte layout to ROOT. The 2024 NanoAOD release of 2016 data is a real step, readable without CMS software, but it too is ROOT-based and holds only reconstructed objects.

Reproducibility as a by-product

Every published result should come with open code and be reproducible. Today CMS guarantees reproducibility only for a few flagship results, such as the Higgs discovery. Two things stand in the way: internal tooling, and the academic view of analysis code as a trade secret to be kept hidden.

Large collaborations like CMS should move away from that view and aim at full reproducibility, but doing so by hand could be very time-consuming. Instead, I propose a new paradigm: analysts work in a controlled environment that records every command and action. The record makes reproducibility a by-product and paves the way for AI-aided reproduction and workflow verification that could spot human errors.

Why low-level and RAW data still matter

High-level data is easier to use, but the field still needs low-level data, up to the RAW detector readout:

1. **Full reproducibility.** A complete pipeline starts with RAW-to-reconstruction. CMS publishes mostly high-level data partly to save space, and cheaper storage weakens that reason.
2. **AI-driven reconstruction and detector design.** Both need detector-level inputs; we needed them enough to regenerate the simulation ourselves.
3. **Realistic testing.** Most public ML benchmarks in particle physics use idealized, simplified samples, and even full simulation is cleaner than real data. Real data is messy: partly uncalibrated, with detector regions off or underperforming and changing conditions. With its data-quality flags, low-level collision data would give a real testbed and a more realistic measure of performance.

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